12.8 Penney's Game

WORKSHEET

The question

STEP III 2001, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s3-q13

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Suppose the tosses so far end in TT and nobody has won yet. Can THH ever appear before TTH?

STAGE 1 States

Covers: Part (i)

Ask yourself Once TT has appeared, can THH ever come before TTH? Then let p_{XY} be the probability that A wins when the last two tosses are XY. Write an equation for each of the four states.

STAGE 2 Solving

Covers: Part (i)

Ask yourself Solve the four equations. Each starting pair has probability $\frac{1}{4}$.

STAGE 3 B against C

Covers: Part (ii)

Pause and try C wants HHT and B wants THH. If the game starts HH, who wins? If it doesn't, look at what comes just before the first HH.

STAGE 4 The cycle

Covers: Part (iii)

Ask yourself For C against D, set up states as in Part (i). For D against A, swap heads and tails in Part (ii).

What's really going on

Why some sequences are better

All sequences of three tosses are equally likely at any given position, and the expected waiting time for each is easy to compute. But the waiting times differ: you wait on average 8 tosses for TTH but 14 for TTT, because a sequence that overlaps itself (like TTT, where a near miss TTH ruins two tosses of progress) is slow to appear. And in a race between two sequences, what matters is how each overlaps the *other*. THH beats HHT because every occurrence of HHT (except at the very start) is preceded by a T, which means THH has just occurred. The second player's trick is always the same: take the first two symbols of your opponent's sequence, and put in front of them the opposite of the opponent's middle symbol.

Conway's formula

John Conway found a remarkable formula for the odds. For two sequences X and Y of length n, define the *correlation* XY as a binary number with n digits: reading from the left, the ith digit is 1 if the part of X starting at its ith symbol equals the beginning of Y of the same length. Then the odds that Y appears before X are

$$(XX - XY) : (YY - YX).$$

For $X = \text{HHT}$ and $Y = \text{THH}$: $XX = 100_2 = 4$ (only the whole of HHT matches itself); $XY = 001_2 = 1$ (only the final T of HHT matches the start of THH); $YY = 100_2 = 4$; and $YX = 011_2 = 3$ (both HH and H at the end of THH match the start of HHT). So the odds that THH comes first are $(4 - 1) : (4 - 3) = 3 : 1$, the $\frac{3}{4}$ of Stage 3. Conway's formula works for sequences of any length and any alphabet, and its proof uses the theory of *martingales*: imagine a casino in which gamblers bet on the sequence appearing, and use the fact that a fair game has zero expected profit.

Penney and Gardner

Walter Penney described the game in a short note in the *Journal of Recreational Mathematics* in 1969. Martin Gardner made it famous in his *Scientific American* column in 1974, including Conway's formula, which Conway had found but not published. Non-transitive relations of this kind (Efron's dice, where die A beats B, B beats C, C beats D and D beats A, are another example) are a recurring source of surprise, and they matter in voting theory, where majority preferences can form cycles: Condorcet's paradox.

Try this next

1. **Four players together.** Try STEP II 2015, Question 12 (step.maths.org/questions/15-s2-q12): Penney's game with four players at once.

2. **Waiting times.** Show that the expected number of tosses until HH first appears is 6, and until HT is 4. (Use first-step analysis.)

3. **The best reply.** For each of the eight sequences of length three, find the sequence that beats it with the greatest probability. Which sequence is beaten most easily?

4. **Conway's formula.** Use Conway's formula to check that HHT beats HTT with probability $\frac{2}{3}$.

Further reading

- **Martin Gardner, *Time Travel and Other Mathematical Bewilderments*** (W. H. Freeman). Includes Penney's game and Conway's formula.
- **Ronald Graham, Donald Knuth & Oren Patashnik, *Concrete Mathematics*** (Addison-Wesley), Section 8.4. Penney's game with generating functions.
- **David Williams, *Probability with Martingales*** (Cambridge University Press). The martingale proof, for later.

The full guided solution is in Section 12.8:
https://beyondstep.keithmcnulty.org/ch12/penney.html